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React JS Lesson 8 – useEffect Hook (Premium Notes)

1) What is useEffect

- useEffect is a React Hook used to perform side effects after rendering.
- It runs after component render completes.

Example:

```
useEffect(()=>{  
  console.log("Component Rendered");  
});
```

Explanation:

- Effect runs after every render when no dependency array is given.

2) Why we use useEffect (Side Effects)

- Used for API calls, timers, local storage, subscriptions.

Example:

```
useEffect(()=>{  
  document.title = "React App";  
});
```

Explanation:

- Browser tab title updates after render.

3) useEffect syntax

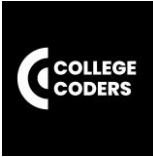
- useEffect takes callback function and optional dependency array.

Example:

```
useEffect(()=>{  
  // logic  
},[]);
```

Explanation:

- First part = effect logic.
- Second part = dependency control.



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4) useEffect without dependency array

- Runs after every render.

Example:

```
useEffect(()=>{  
  console.log("Runs every render");  
});
```

Explanation:

- Every state change triggers effect again.

5) useEffect with empty dependency array

- Runs only once after first render.

Example:

```
useEffect(()=>{  
  console.log("Runs only once");  
},[]);
```

Explanation:

- Useful for first load setup.

6) useEffect with dependency array

- Runs when dependency changes.

Example:

```
useEffect(()=>{  
  console.log(count);  
},[count]);
```

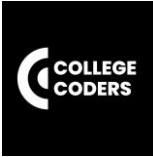
Explanation:

- Effect runs only when count changes.

7) Cleanup function inside useEffect

- Cleanup removes timers or subscriptions.

Example:



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```
useEffect(()=>{  
  const timer = setInterval(()=>{},1000);  
  
  return ()=>{  
    clearInterval(timer);  
  }  
},[]);
```

Explanation:

- Return function cleans previous effect.

8) Unmounting & cleanup on component removal

- Cleanup runs when component disappears.

Example:

```
return ()=>{  
  console.log("Component Removed");  
}
```

Explanation:

- Useful to avoid memory leaks.

9) Practical example – input watcher

- Dependency tracks input changes.

Example:

```
useEffect(()=>{  
  console.log(name);  
},[name]);
```

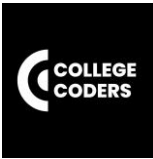
Explanation:

- Runs only when name changes.

10) Mini Project – Auto Timer + Stop Timer

- Timer increases every second.
- Cleanup stops timer safely.

Example:



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```
useEffect(()=>{  
  const timer = setInterval(()=>{  
    setCount(prev=>prev+1);  
  },1000);  
  
  return ()=>clearInterval(timer);  
},[]);
```

Explanation:

- setInterval increases count.
- clearInterval stops timer.

Assignment

- Create timer app.
- Add start and stop buttons.
- Use cleanup function correctly.